Rafael A. Rodriguez-Sanchez

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About

Ph.D. candidate in Computer Science with research spanning representation learning, reinforcement learning, and hierarchical RL. My work leverages energy-based and contrastive methods to improve state representations in RL. Proficient in Python and experienced in developing deep learning algorithms in PyTorch and JAX.

Education

Brown University — **Providence, RI**
Ph.D. in Computer Science; — Dec 2025 (Expected)
Dissertation: "Action-driven Algorithms for Representation Learning in Reinforcement Learning"
Advisor: George Konidaris

Politecnico di Milano — **Milan, Italy**
M.S. in Computer Science and Engineering, *Cum Laude* — 2018
Thesis: "A Variational Approach to Transfer Value Functions in Reinforcement Learning"
Advisor: Marcello Restelli

Universidad Simon Bolivar — **Caracas, Venezuela**
B.S. in Electronic Engineering, *Summa Cum Laude* — 2016

Experience

Amazon (Alexa), Applied Scientist Intern — **Cambridge, MA** | Summer 2021
Finetuned pretrained LLMs (via SFT and RL) for Semantic Parsing in task-oriented dialog systems.

Politecnico di Milano, Research Fellow, AIRLab — **Milan, Italy** | Fall 2018–Summer 2019
Developed vision-based tracking algorithms for intelligent missiles; simulation to support optimal defense strategies.

STMicroelectronics, R&D Intern — **Agrate-Brianza, Italy** | Feb –Jul 2015
Developed real-time algorithm to select optimal image-acquisition timing; improved acquisition throughput; debugged power and image transmission on a microcontroller board.

Conference Publications

- **R. Rodriguez-Sanchez**, C. Allen, G. Konidaris. *From Pixels to Factors: Learning Independently Controllable State Variables for Reinforcement Learning. Under review 2025.* [preprint]
- A. Bagaria, A. De Mello Koch, **R. Rodriguez-Sanchez**, S. Lobel, G. Konidaris. *Intrinsically Motivated Discovery of Temporally Abstract Graph-based Models of the World*. RLC 2025.
- **R. Rodriguez-Sanchez**, G. Konidaris. *Learning Abstract World Models for Value-preserving Planning with Options*. RLC 2024.
- **R. Rodriguez-Sanchez***, B. Spiegel*, J. Wang, R. Patel, S. Tellex, G. Konidaris. *RLang: A Declarative Language for Describing Partial World Knowledge to Reinforcement Learning Agents*. ICML 2023.
- **R. Rodriguez-Sanchez***, A. Tirinzoni*, M. Restelli. *Transfer of Value Functions via Variational Methods*. NeurIPS 2018.

Workshops

- **R. Rodriguez-Sanchez**, C. Allen, G. Konidaris. *Disentangling Independently Controllable Factors in RL*. NYRL Workshop 2025 (extended abstract).
- **R. Rodriguez-Sanchez**, G. Konidaris. *Learning Abstract World Models for Value-preserving Planning with Options*. GenPlan @ NeurIPS 2023 ***[Oral]***
- **R. Rodriguez-Sanchez***, R. Patel*, G. Konidaris. *On the Relationship Between Structure in Natural Language and Models of Sequential Decision Processes*. LaReL @ ICML 2020.

Skills & Relevant Coursework

Python, **JAX, PyTorch**
Machine Learning, Deep Learning, Advanced Topics in Vision and Language, Algorithmic Game Theory, Optimization, Sequential decision-making.

Languages

Spanish (native) | English (C2 Level) | Italian (C1 Level)

Scholarships and Awards

MAECI Scholarship (Italian Ministry of Foreign Affairs) — full Master's tuition and living expenses 2017–2018
USB "Exceptionally Good" Mention for Undergraduate Thesis 2015
USB Top 30 Students across all majors 2016
USB Best Electronic Engineering Student (Cohort 2010) 2012, 2014
USB Top 10 Students of 2010 Cohort 2011